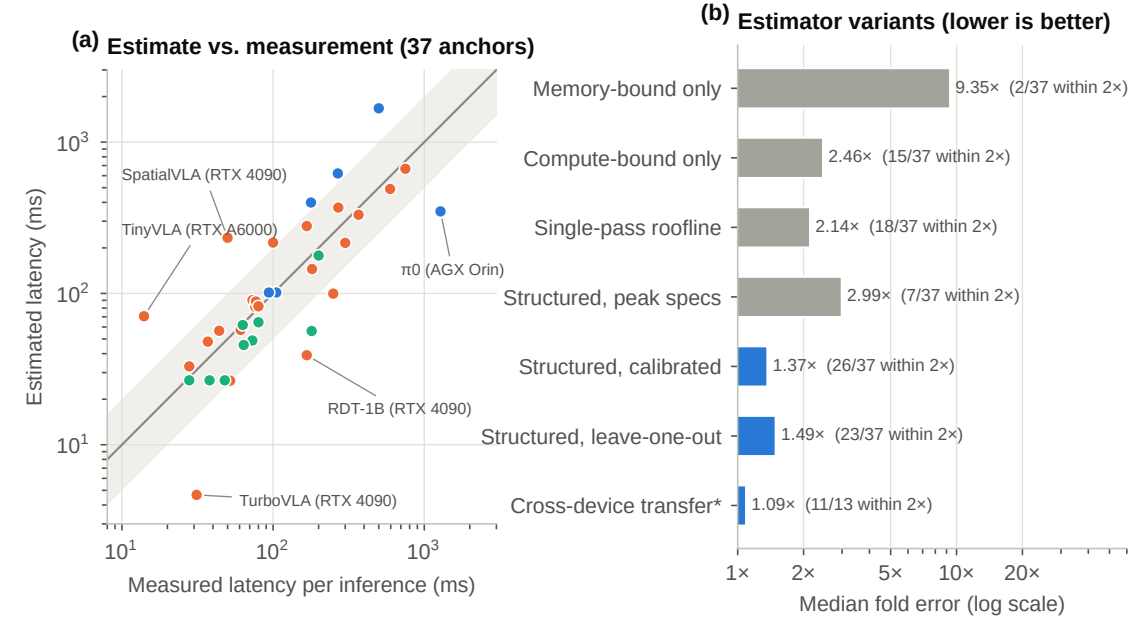
(a) Estimate vs. measurement (37 anchors)
Estimated latency (ms)
Measured latency per inference (ms)
10^1
10^2
10^3
SpatialVLA (RTX 4090)
TinyVLA (RTX A6000)
π0 (AGX Orin)
RDT-1B (RTX 4090)
TurboVLA (RTX 4090)
(b) Estimator variants (lower is better)
Memory-bound only
9.35× (2/37 within 2×)
Compute-bound only
2.46× (15/37 within 2×)
Single-pass roofline
2.14× (18/37 within 2×)
Structured, peak specs
2.99× (7/37 within 2×)
Structured, calibrated
1.37× (26/37 within 2×)
Structured, leave-one-out
1.49× (23/37 within 2×)
Cross-device transfer*
1.09× (11/13 within 2×)
1×
2×
5×
10×
20×
Median fold error (log scale)
Within 2×
Embedded (Jetson) (n = 6)
Desktop / workstation GPU (n = 22)
Data-centre GPU (n = 9)